

Classification of EEG with Structural Feature Dictionaries in a Brain Computer Interface

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Abstract— We present a new method for the classification of EEG in a brain computer interface by adapting subject specific features in spectral, temporal and spatial domain. For this particular purpose we extend our previous work on ECoG classification based on structural feature dictionary and apply it to extract the spectro-temporal patterns of multichannel EEG recordings related to a motor imagery task. The construction of the feature dictionary based on undecimated wavelet packet transform is extended to block FFT. We evaluate several subset selection algorithms to select a small number of features for final classification. We tested our proposed approach on five subjects of BCI Competition 2005 dataset- IVa. By adapting the wavelet filter for each subject, the algorithm achieved an average classification accuracy of 91.4%. The classification results and characteristic of selected features indicate that the proposed algorithm can jointly adapt to EEG patterns in spectro-spatio-temporal domain and provide classification accuracies as good as existing methods used in the literature.

I. INTRODUCTION

A BRAIN computer interface is a system that converts patterns recorded from brain into control and communication signals. Since this system directly uses the activity of brain, it has great potential in helping handicapped people suffering from spinal cord injury or amyotrophic sclerosis. Electroencephalogram (EEG) is frequently used as a modality to record neural activity since it is noninvasive, inexpensive and has minimal clinical risks. In particular motor imagery related EEG patterns have been used in construction BCIs. In these types of systems the subjects are asked to perform an imagination of a particular motor activity such as hand or foot movement. Related oscillatory patterns are extracted from central areas for classification and used a control command or as a feedback to the user. In this framework signal processing methods such as autoregressive modeling [1] and common spatial patterns [2] are widely preferred as a front end for feature extraction. The adaptive autoregressive modeling has highlighted the importance of temporal adaptation to

spontaneous EEG patterns to represent the dynamic spectrum and was used to estimate the optimal time point for classification [1]. In the following stage the common spatial patterns was introduced for spatial adaptation by manually selecting an active time window and the relevant subbands of EEG [2]. Recently, the research in this area has been extended into adaptive time-frequency analysis of EEG for the extraction of multiscale features in a BCI [3]. All of these steps have emphasized the importance of adaptation to EEG patterns in space, time, and frequency. In this paper we present an approach that can accomplish this task by constructing a rich structural feature dictionary that extracts spectro-spatio-temporal patterns of multichannel EEG associated to movement imagery tasks. We describe two different main methods of constructing the structural feature dictionary based on (i) subband filtering and (ii) block transform. In our previous study [4] we used undecimated wavelet packet transform to decompose the neural activity into subbands in a tree structure and segmented each subband into time segments. In this paper we extend our method to Block Fourier (BF) transform. Basically we substitute spectral and temporal segmentation processes. We use two subset selection algorithms to choose a small number of features for final classification and compare the efficiency of the extracted features with wavelet transform and block FFT. The subset selection procedure evaluates a combination of features that originate from different spectral indices, temporal and/or spatial locations. The final selected subset of features is fed to the classifier for final decision.

The paper is organized as follows. In the next section we describe filtering and block transform based methods for structural feature dictionary design. In section III, we summarize the algorithms used to select an efficient subset of features for final classification. In section IV we describe the EEG data set and provide experimental results. Section V constitutes the conclusions.

II. FEATURE GENERATION

A. Undecimated Wavelet Packet Transform

The main structure in generating features is a dual tree as shown in Figure 1. Basically, we want to exploit the time and frequency content of the signal at a desired resolution. Therefore in our previous studies we decomposed the neural activity into subbands. Furthermore, each subband is divided into time segments. Both time and frequency segmentation steps use a dyadic tree structure. In our previous work, we used Undecimated Wavelet Packet Transform (UDWPT) to

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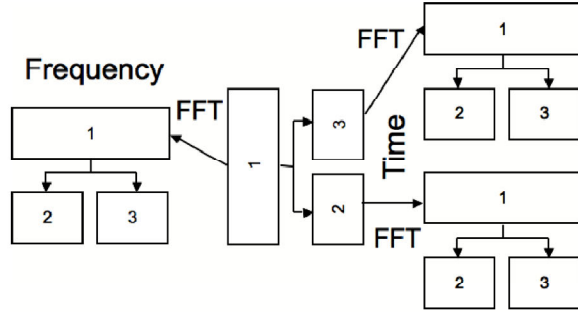


Figure 1. Dual Tree Structure using the Block Fourier Transform.

extract the subband activity. The undecimated wavelet transform is adapted to have the final transform to be shift invariant [5] which is crucial in a pattern recognition framework. Once the segmentation is completed in time and frequency axis, the energy in each time segment, sum of coefficients' squares, is kept as one feature. This constitutes a feature space with dimension $N = (2^{(T+1)} - 1) (2^{(F+1)} - 1)$ where T is the time level and F is the frequency level. The reader is referred to [4] for details.

B. Block Fourier Transform

In this paper we propose to use the block fast Fourier Transform (BF) in addition to the WP Transform to generate the features. The features are generated using the same dual tree structure explained in the previous part. The only difference is that the subbands are extracted using harmonics. However, in this process the time segmentation precedes the frequency segmentation.

First, as explained in the previous section, the original signal is divided into $(2^{(T+1)} - 1)$ dyadic time blocks. Then, for each time block an N -point FFT is calculated, N being the length of the signal. Finally, each N -point FFT is segmented into $(2^{(F+1)} - 1)$ dyadic segments to extract the subbands. This is accomplished by merging consecutive expansion coefficients in desired subbands. The energy in each subband is kept as one feature. As in the previous case, there will be total $(2^{(T+1)} - 1) (2^{(F+1)} - 1)$ features.

The feature generation algorithm using the BF transform is explained following the steps below using an example provided in Figure 1 for two levels both in time and frequency. Hence, the total number of features is $3 \times 3 = 9$.

Step 1: Divide the original signal into three dyadic time blocks, the vertical blocks numbered as 1, 2, and 3 in the middle.

Step 2: Calculate the FFT for each time block, horizontal blocks numbered as 1. This will make the first subbands.

Step 3: Divide each subband in Step 2 into two more dyadic segments to form the additional subbands, horizontal blocks numbered 2 and 3.

Step 4: Calculate the energies of 9 subbands extracted above to form the features.

Compared to wavelet based method we note that block transform based method requires less memory and resulting

features are faster to compute. In addition, transition effects on the borders of the signal as in wavelet transform do not exist in the block FFT case. Finally, in the wavelet case the selection of mother wavelet is another free parameter that one needs to deal with.

III. SUBSET SELECTION

The motivation in the feature dictionary design using one of the methods explained in the previous section is to generate a redundant feature dictionary and let the classifier select a subset of them. This way we will have a computationally inexpensive system which is desired. However, in a high dimensional feature space without regularization the classifier will overfit easily; hence, a subset selection is required. To select a subset of features we employ a wrapper method and a filtering method. In these algorithms, the features are ordered based on a particular cost function, and then a number of them are selected using cross validation. These selected features are fed to the classifier. The proposed classification approach with subset selection is shown in Figure 2.

The cost function to evaluate the discrimination ability of the features is the Fisher Discrimination (FD) criterion. It measures how well two distributions are separated from each other in a projected space [6]. In equation (1) μ_i and σ_i^2 represent the mean and variance of i^{th} distribution, respectively.

$$FD = (\mu_1 - \mu_2)^2 / (\sigma_1^2 + \sigma_2^2) \quad (1)$$

A. Sequential forward feature selection with pruning (SFFS-P)

This procedure implements a wrapper strategy to sequentially select features. In addition, SFFS-P executes a pruning method by using the structural relationship between dual tree nodes. In particular the pruning uses the information that all features have a known location in time and frequency. When a feature is selected using the SFFS, the features that overlap in both the time and frequency are discarded. The SFFS algorithm is then applied on the remaining features. This reduces the computational complexity when compared to the stand alone sequential forward selection procedure. Note that the SFFS-P procedure evaluates the efficiency of the combination of features in the structural dictionary

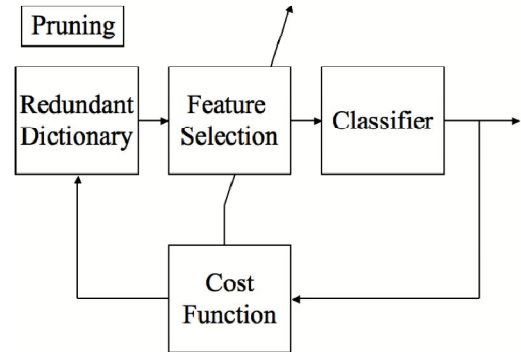


Figure2. The block diagram of the proposed classification method.

B. Cost function based pruning (CFS)

The CFS is a basic filtering technique based on the FD criterion to sort all the features individually. The feature with the maximum discrimination is selected and then all other features with the overlapping time and frequency are eliminated with the pruning procedure. This step is iterated until there are no more features left for pruning. This approach does not use the feedback in Figure 2. After this filtering is done, the features are sorted by their discrimination power. Since no feedback is used the computational cost is lesser than SFFS-P.

C. Multi Channel EEG Data

To evaluate the performance of the algorithm we used the EEG dataset IVa of BCI competition 2005 [7] recorded from 5 subjects (**aa**, **al**, **av**, **aw**, **ay**). The subjects were asked to imagine either foot or right index finger movement. The EEG data were recorded with 1 kHz sampling rate from 118 electrodes. In order to reduce the size of the data, these signals were filtered with a low pass filter in the 0-40 Hz band and then down-sampled to 125Hz. In this study the EEG data from 33 electrodes over the motor and central areas are analyzed. These channels are converted to Hjort derivation in order to enhance the local activity [8]. We used a time window of 4 seconds following the go signal for motor imagery. There are 140 trials from each class. The classifier used in this study is a linear discriminant classifier.

IV. RESULTS

In this study we set both the time and frequency levels to 4 when constructing the dual tree. Given the duration of the EEG signal being 4 seconds, this corresponds to a finest time resolution of around 250ms. And due to 64 Hz signal bandwidth the finest frequency resolution is around 4 Hz. In the feature generation process using the dual tree we have one last free parameter to decide that is the basis functions. When the BF transform is used the basis functions are fixed. With the WP transform there are many filters to choose from. In order to see the effect of vanishing moments on overall classification performance, three different wavelet filters are used; namely, 8 tap dB4, 12 tap dB6, and 16 tap dB8 Daubechies filters, respectively.

In order to evaluate the prediction performance of the binary BCI classification problem, a double resampling method is employed. A 7-fold cross validation method is used to estimate the test error and 10-fold cross validation method

is used to estimate the optimal model complexity (number of features) for each training fold. The estimated prediction performance is provided as the average of seven test errors. In Table I we provide the best results obtained by WP based features and BF based features for each subject. The average number of features (NoF) used to reach those test errors is also provided. In addition to these results, the rightmost two columns show the results obtained using Common Spatial Patterns (CSP) and Autoregressive (AR) models based

TABLE I

THE CLASSIFICATION PERFORMANCE (%) OF THE PROPOSED ALGORITHM AND. THE NUMBER OF FEATURES (NOF) TO REACH MINIMAL ERROR IS PRESENTED. THE RESULTS OF CSP AND AR METHODS ARE ALSO GIVEN FOR COMPARISON.

Subject	WP		BF		CSP	AR
	Test Error	NoF	Test Error	NoF	Test Error	Test Error
<i>aa</i>	9.3	33.6	12.1	4.1	8.5	10.5
<i>al</i>	1	6.3	2.1	4.1	0.8	1.6
<i>av</i>	23.6	22.3	23.6	22.9	29.1	23.3
<i>aw</i>	3.6	47.6	5	30.6	3.1	7.7
<i>ay</i>	5.4	6.6	6.8	5.4	5.3	9.5
<i>Avg.:</i>	8.6	23.3	9.9	13.4	9.4	10.5

approaches [9]. In that study Support Vector Machine (SVM) was used as the classifier in a 10 fold cross validation experiment.

Between the dual tree structure based methods the best classification performance is achieved with the WP based features. It achieved lower classification errors in four of the five subjects, and achieved the same on the remaining subject, *av*. But it should be noted that the average number of features is less with BF based features. When compared with the CSP and AR based approaches in terms of average test errors on all subjects, we observe that the results obtained with dual tree structure are better than that of AR based approach and comparable to that of CSP based approach. But it should be noted that the classifier used with the latter methods was an SVM. Since in this study our main aim was to explore the effect of features on the final classification performance no special attention is given to the classifier and only the simple LDA classifier is considered. It is known that the LDA is optimal when the distribution in the input domain is Gaussian whereas the SVM is optimized for finite data setting [10]. Hence, in this finite sample problem setting SVM has an advantage over LDA. However due to the complexity associated with its training, using SVM in SFFS-P procedure is infeasible.

The main characteristics of the results with different subset selection algorithms are that the minimum error is achieved with a few numbers of features with SFFS-P. On the other hand, CFS uses more features to reach the minimum error. The results provided in Table I are obtained with the following algorithms. On subject *aa*, WP achieved the best result using CFS and BF algorithm with SFFS-P. On the remaining subjects the best results are achieved using the same algorithms; SFFS-P, CFS, CFS, and SFFS-P, respectively. We believe the reason behind achieving the best results with the same types is due to the same dual tree structure was used in both WP and BF based feature generation. The variability in the number of features used to reach the minimum test results provided in the Table I can now be understood with respect to the algorithms they use.

As mentioned earlier we evaluated three different wavelet filters to see their effects on the final classification performance. The best results on each subject are obtained with dB8, dB4, dB8, dB4, and dB6 wavelet filters, respectively. Clearly, we can not say a priori that the

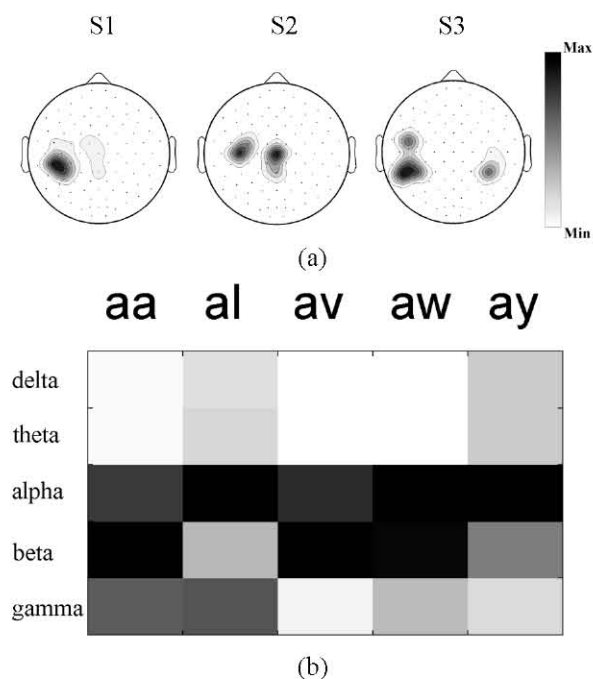


Figure 3. (a) Spatial distribution of selected features for subjects aa, al and av. We note distinct regions were selected for each subject. (b) The distribution of selected subbands of each subject. Note that in general alpha and beta frequency bands are selected by our algorithm.

features generated by a certain filter may perform better than the others. The reason for this behavior we believe is that for the same subject and type the feature order is different for distinct filters. Additionally, in order to quantify the differences in final classification accuracies we run paired t-tests between classification results obtained by WP and those of by BF for a fixed filter and feature selection algorithm over all subjects. When the feature selection algorithm is the same and the wavelet filter is fixed, the results between the two methods do not differ significantly at $p=0.05$ level. In contrast, if the wavelet filter that gives the best result for each subject is selected in each feature selection algorithm, then the classification results do differ significantly ($p\text{-val}=0.041$). This suggests that the structural dictionary generated with BF provides as good results as that of based on WP with the fixed wavelet filter. The overall superiority of WP based method comes from subject adapted wavelet filters.

In order to better understand physiological basis that provided best classification results we extracted subband indices of selected features and corresponding electrode positions. We counted the number of features located in δ -band (1-4 Hz), θ -band (4-8 Hz), α -band (8-16 Hz), β -band (16-32 Hz), and γ -band (32-64 Hz), respectively. We also calculated the number of features from each electrode position. These counts are mapped over the bands and over the channels and depicted in Figure 3. In order to be able to make a comparison the counts are max-normalized. The characteristic of selected features indicate that the proposed algorithm selected subject specific EEG patterns in

spectro-spatial domain.

V. CONCLUSIONS

In this study we extended the recently proposed dual tree structure based feature generation. Namely, in addition to using wavelet filters to generate the structural time frequency feature dictionary we also employed Block Fourier transform. The biggest difference between these two feature generations is that there is no filter component to choose when using the Block Fourier transform. The feature sets generated by WP and BF are used in a binary classification learning problem setting targeted for an EEG based BCI task. We obtain that the classification performances of the dual tree structure based features are comparable to that of CSP based approach and better than that of the AR based approach reported on the same data. When for each subject the wavelet filter that provides the best test result is selected, on average the performance of the WP based features are better than that of BF based features. By adapting the wavelet filter for each subject a better performance is available with and an increased computational cost. On the other hand, the average number of features used by BF method is almost half compared to the WP based method. The overall learning method does not depend on a priori spatial information, temporal and spectral content of the EEG signals under investigation. Rather, it selects the most discriminant space-time-frequency features from the dictionary in a subject adaptive manner.

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